b) Find the sum of the series 3

$$1 + x\cos\theta + x^2\cos 2\theta + \cdots + x^{n-1}\cos(n-1)\theta$$

c) Separate into real and imaginary parts of the expression $\tan^{-1}(x+iy)$. 2.75

6)

a) Show that the vectors $\vec{A} = 3\vec{\imath} - 2\vec{\jmath} + k, \vec{B} = \vec{\imath} - 3\vec{\jmath} + 5\vec{k}, \vec{C} = 2\vec{\imath} + \vec{\jmath} - 4\vec{k}$ form a right angled triangle. 3

b) Prove that the area of a parallelogram with sides $\underline{A}$ and $\underline{B}$ is $|\underline{A}| \times |\underline{B}|$. 2.75

c) Prove that $\nabla \times \nabla\phi = 0$ and $\nabla.(\nabla \times A) = 0$ 3

7)

a) If $A = (3x^2 + 6y)i - 14yzj + 20xz^2k$, evaluate $\int_C A.dr$ from (0,0,0) to (1,1,1) along path $C: x = t, y = t^2, z = t^3$. 3

b) Evaluate $\iint \varphi n ds$, where $\varphi = \frac{3}{8}xyz$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$. 3

c) Find the volume of the region common to the intersecting cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$. 2.75

8)

a) Verify Green's theorem in the plane for $\oint_C (xy + y^2)dx + x^2dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. 3

b) If $v = w + r$, prove that $w = \frac{1}{2}Curl\ V$, where w is a constant vector. 3

c) Evaluate $\iint_s F.nds$, where $\vec{F} = 4x\vec{\imath} - y^2\vec{\jmath} + yzk$ and S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$. 2.75

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(b) Show that for the curve $by^2 = (x+a)^2$, the square of the subtangent varies as the subnormal. 3

(c) Prove that the asymptotes of the cubic $(x^2-y^2)y - 2ay^2 + 5x - 7 = 0$ form a triangle of area a^2. 3

Section-B

5. (a) Evaluate any three of the following integrals: 3.75

(i) $\int \frac{e^x}{x}(1+xlogx)dx$ (ii) $\int \frac{x}{(x-1)(x-2)}dx$

(iii) $\int \frac{dx}{5+4cosx}$ (iv) $\int \frac{x}{x+\sqrt{(x^2-1)}}dx$

(b) What is the geometrical interpretation of $\int_a^b f(x)dx$? Evaluate from first principle $\int_0^1 x^2\,dx$. 3

(c) Evaluate $\lim_{n\to\infty}\{\frac{n}{n^2+1^2}+\frac{n}{n^2+2^2}+\frac{n}{n^2+3^2}+\cdots+\frac{n}{n^2+n^2}\}$. 2

6. (a) Evaluate the following definite integrals: 4

(i) $\int_8^{15} \frac{dx}{(x-3)\sqrt{(x+1)}}$ (ii) $\int_1^{e^2} \frac{dx}{x(1+logx)^2}$

(b) Show that $\int_0^{\frac{\pi}{2}} \frac{1}{(a^2cos^2x+b^2sin^2x)}dx = \frac{\pi}{2ab}$. 1.75

(c) Use necessary properties of definite integrals to evaluate the following integrals mentioning each of the properties you used in the evaluation process: 3

(i) $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x+\cos x}dx$ (ii) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(\sin x)^{51}\,dx$

7. (a) What do you mean by a reduction formula? Obtain a reduction formula for $\int tan^n x dx$ and hence find the value of $\int_0^{\frac{\pi}{4}} tan^4 x dx$. 3

(b) Define Beta and Gamma functions. Mention their applications. Show that 3

$$\int_0^{\frac{\pi}{2}} sin^p\theta cos^q\theta d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}, p > -1, q > -1.$$

(c) Evaluate the integral $\int_0^{\infty} e^{-x^2}dx$ with the aid of Gamma function. 2.75

8. (a) Find, by integration, the area between the curve $y = x^2 - 4$ and the x–axis. 3

(b) Use the concept of integrals to find the perimeter of the circle $x^2 + y^2 = a^2$. 2.75

(c) Find the volume of the solid generated by revolving the cardioid $r = a(1+cos\theta)$ about the initial line. 3

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Section-B

5. (a) Separate $\log(a+ib)$ into real and imaginary parts. 3

(b) If $\tan(x+iy) = u+iv$, prove that $u^2+v^2+2u\,cot2x = 1$. 2.75

(c) If $x = \frac{2}{1!} - \frac{4}{3!} + \frac{6}{5!} - \frac{8}{7!} + \cdots to\ \infty$ and $y = 1 + \frac{2}{1!} + \frac{2^3}{3!} - \frac{2^5}{5!} - \cdots to\ \infty$, then show that $x^2 = y$. 3

6. (a) Sum the series $cos\alpha + cos2\alpha + cos3\alpha + \cdots + cos n\alpha$. 3

(b) Prove that $i^i = e^{-(4n+1)^{\pi/2}}$. 2.75

(c) If $\tan(x+iy) = u+iv$, prove that $u^2 + 2u\,cot2x + v^2 = 1$. 3

7. (a) If $\vec{A}$ is any vector, then prove that $\vec{A} = (\vec{A}.\hat{\imath})\,\hat{\imath} + (\vec{A}.\hat{\jmath})\,\hat{\jmath} + (\vec{A}.\hat{k})\,\hat{k}$. 2.75

(b) Let the position vector of the points P and Q are $r_1 = 2\vec{\imath} + 3\vec{\jmath} - \vec{k}$ and $r_2 = 4\vec{\imath} - 3\vec{\jmath} + 2\vec{k}$. Determine the vector PQ and its magnitude. 3

(c) If $\vec{A} = 4\vec{\imath} - \vec{\jmath} + 3\vec{k}$ and $\vec{B} = -2\vec{\imath} + \vec{\jmath} - 2\vec{k}$, Find $A \times B$. 3

8. (a) Find the angle between the surfaces $x^2+y^2+z^2 = 9$ and $z = x^2+y^2-3$ at the point (2,-1,2). 3

(b) State and prove Green's theorem. 2.75

(c) Find the value of $\int_{-3}^{3}\int_{0}^{4}\int_{2}^{5}(x+y+z)dx\,dy\,dz$.

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7. a) Define Beta and Gamma functions. Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. 3.5

b) If $u_n = \int_0^{\pi/2} x^n \sin x \, dx$, then show that $u_n + n(n-1)u_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}$. 3.5

c) Evaluate the integral $\int_0^{\infty} e^{-x^2} \, dx$ with the aid of Gamma function. 3

8. a) Find the area bounded by the curves $y^2 = 4ax$ and $x^2 = 4ay$. 3

b) Find the whole length of the loop of the curve $3ay^2 = x(x-a)^2$. 3.5

c) The circle $x^2 + y^2 = a^2$ revolves round the x-axis. Show that the surface area and the volume of the whole sphere generated are respectively $4\pi a^2$ and $\frac{4}{3}\pi a^3$. 3.5

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7. (a) Define Beta and Gamma functions. Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. 3

(b) If $I_n = \int_0^{\frac{\pi}{4}} tan^n\theta \, d\theta$, then show that $n(I_{n+1} + I_{n-1}) = 1$. 3

(c) Evaluate the integral $\int_0^{\infty} e^{-x^2} \, dx$ with the aid of Gamma function. 2.75

8. (a) Find the perimeter of the circle $x^2 + y^2 = 16$ using the concept of integrals. 2.75

(b) Find the area of the quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ between the major and minor axes. 3

(c) If the circle $x^2 + y^2 = a^2$ revolves round the $x - axis$, then find the volume of the whole sphere generated. 3

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S-A

University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2021
Course: MATH1121 (Differential and Integral Calculus)
(According to the Syllabus:2019-2020)
Marks: 52.50 Time: 3 Hours

(Answer any THREE questions from each Section)

Section-A

1. (a) Find the domain and range of the function $f(x) = [x]$, also draw the graph of this function. 3
   (b) Define limit of a function $f(x)$ at $x = c$. Find the value of $\lim_{x\to 0} f(x)$ where $f(x) = |x|$. 2.75
   (c) Using the L'Hospital's theorem evaluate $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x}}$. 3

2. (a) Define the continuity and differentiability of a function $f(x)$ at a point $x = a$. Show that if a function $f(x)$ is differentiable at $x = a$ then it must be continuous at $x = a$. 3
   (b) Differentiate $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ with respect to $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$. 2.75
   (c) Define the continuity of a function $f(x)$ at a point $x = c$. Discuss the continuity of 3

   The function $f(x) = \begin{cases} \frac{1}{2} - x & if\ 0 < x < \frac{1}{2}, \\ \frac{1}{2} & if\ x = \frac{1}{2} \\ \frac{3}{2} - x & if\ \frac{1}{2} < x < 1 \end{cases}$ $at\ x = \frac{1}{2}$.

3. (a) If $y = \sin(m\sin^{-1}x)$, then using Leibnitz theorem prove that 3
   $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$.
   (b) Find the maximum and minimum values of z where $z = \frac{4}{x} + \frac{36}{y}$ and $x + y = 2$. 3
   (c) Find the value of θ in the mean theorem $f(x+h) = f(x) + hf'(x+\theta h)$, where $f(x) = \frac{1}{x}$. 2.75

4. (a) Prove that the largest rectangle with a given perimeter is a square. 2.75
   (b) State Euler's theorem on homogeneous function. If $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, show that 3
   $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$.
   (c) Find two positive numbers whose sum is 100 and the sum of whose square is minimum. 3

Section-B

5. (a) Evaluate the following integrals: 4
   (i) $\int \frac{\cot x}{\log \sin x} dx$ (ii) $\int \frac{dx}{5+4\sin x}$
   (b) Write down the geometrical interpretation of $\int_a^b f(x)dx$. 1.75
   (c) Evaluate $\lim_{n\to\infty}\left\{\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \cdots + \frac{n}{n^2+n^2}\right\}$. 3

6. (a) Evaluate $\int_a^b e^x dx$ from the definition of definite integral. 3
   (b) Evaluate $\int_0^1 x^3(1-x^2)^{\frac{5}{2}}\ dx$ 3
   (c) Evaluate $\int_0^{\frac{\pi}{2}} x\log(\sin x)dx$ 2.75

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc.in Engineering 1st Year 1st Semester Examination-2019
Course: **MATH 1211** [Differential and Integral Calculus]
Marks: 52.50 Time: 3 Hours
[Answer any six (06) questions taking three (03) from each section.]

Section-A

1.a) Define function and hence define domain codomain with an example. 3
b) Find the domain and range of $f(x) = |x+1| + |x|$ and also sketch the graph of $f(x)$. 3
c) Prove that every differential function is continuous but the converse is not true. 2.75

2.a) Define derivative of a function $f(x)$ at $x = c$. Show that the function 4

$$f(x) = \begin{cases} 3 + 2x \; for -\frac{3}{2} < x \leq 0 \\ 3 - 2x \; for \; 0 < x \leq \frac{3}{2} \end{cases}$$ is continuous at $x = 0$ but not differentiable at $x = 0$.

b) If $y = (\sin^{-1} x)^2$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$. 3
c) i) Evaluate $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{\frac{1}{x}}$ ii) Differentiate $x^{\sin^{-1} x}$ with respect to $\sin^{-1} x$. 1.75

3.a) For a given curved surface of a right circular cone when the volume is maximum, show that the semi-vertical angle is $\sin^{-1}\frac{1}{\sqrt{3}}$. 4
b) Give the geometrical interpretation of Mean value theorem. 3
c) Verify Rolle's theorem for the function $f(x) = x^2 - 5x + 8$ in $[1, 4]$. 1.75

4.a) If $u = f(x^2 + 2xyz, y^2 + 2zx)$, prove that $(y^2 - zx)\frac{\partial u}{\partial x} + (z^2 - xy)\frac{\partial u}{\partial y} + (x^2 - yz)\frac{\partial u}{\partial z} = 0$. 4
b) Define pedal equation of a curve. Prove that the curve $\frac{x^2}{a} + \frac{y^2}{b} = 1$ and $\frac{x^2}{a'} + \frac{y^2}{b'} = 1$ will cut orthogonally if $a - b = a' - b'$. 3
c) Find the asymptotes of the curve $x^2y^2 - a^2(x^2 + y^2) = 0$. 1.75

Section-B

5. Answer any three of the following: 8.75
i) $\int \frac{x+1}{3+2x-x^2} dx$ ii) $\int \frac{dx}{\sqrt{(2x^2+3x+4)}}$
iii) $\int \frac{dx}{5+4\cos x}$ iv) $\int \frac{dx}{\sqrt{(x^2-2x+3)}\,(x^2-2x+1)}$

6.a) Evaluate $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$. 4
b) Evaluate $\int_2^4 \frac{x-3}{\sqrt{(5-x)(x-1)}} dx$. 3
c) Write down the five general properties of definite integral. 1.75

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University of Rajshahi

(Affiliated College)

Department of Computer Science and Engineering

B. Sc. (Engg.) Part-1 Odd Semester Examination 2021

Course: MATH-1121 (Differential and Integral Calculus)

Full Marks: 60 Duration: 3(Three) Hours

Answer 06 (Six) questions taking any 03 (Three) questions from each Section in separate answer script

Section - A

1. a) Define domain and range of a function. Find the domain and range of the function f defined by $f(x) = \frac{x-3}{2x+1}$. 3.5

   b) A function $f(x)$ is defined as $f(x) = \begin{cases} x & ; 0 \le x < 1/2 \\ 1-x & ; 1/2 \le x < 1 \end{cases}$ 3.5

   Does the function continuous at $x = 1/2$. Also verify the $f(x)$ is differentiable at $x = 1/2$ or not.

   c) Evaluate $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{1/x}$. 3

2. a) Define the continuity of a function $f(x)$ at a point $x = c$. Discuss the continuity of the function $f(x) = \begin{cases} \frac{1}{2} - x & \text{if } 0 < x < \frac{1}{2} \\ \frac{1}{2} & \text{if } x = \frac{1}{2} \\ \frac{3}{2} - x & \text{if } \frac{1}{2} < x < 1 \end{cases}$ at $x = \frac{1}{2}$. 3.5

   b) If $\sin y = x\sin(a+y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$. 3.5

   c) Differentiate $x^{\sin^{-1}x}$ with respect to $\sin^{-1}x$. 3

3. a) State Leibnitz's theorem. If $y = \sin^{-1}x$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$. 3.5

   b) State mean value's theorem. Find the value of c in the mean value theorem $f(b) - f(a) = (b-a)f'(c)$ if $f(x) = x,\ a = 1,\ b = 2$. 3

   c) Find the maximum and minimum values of z where $z = \frac{4}{x} + \frac{36}{y}$ and $x + y = 2$. 3.5

4. a) Prove that the largest rectangle with a given perimeter is a square. 3

   b) State Euler's theorem on homogeneous function. If $u = \cos^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial x} = \frac{1}{2}\cot u$. 3.5

   c) Find the condition that the conics $ax^2 + by^2 = 1$ and $a_1x^2 + b_1y^2 = 1$ shall cut orthogonally. 3.5

Section – B

5. a) Evaluate the following integrals: 5

   i) $\int \sqrt{4 + 8x - 5x^2}\,dx$ ii) $\int \frac{dx}{5 - 13\sin x}$

   b) Write down the geometric interpretation of $\int_a^b f(x)\,dx$. 2

   c) Evaluate $\lim_{n\to\infty}\left\{\frac{1}{1+n} + \frac{1}{2+n} + \cdots + \frac{1}{n+n}\right\}$. 3

6. a) State the Fundamental theorem of Integral calculus. Evaluate $\int_0^a \sqrt{a^2 - x^2}\,dx$. 3.5

   b) Prove that $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx = \pi/4$. 3

   c) Evaluate the definite integral $\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$ 3.5

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University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engineering **Part-I Odd** Semester Examination **2020**

Course Code: **MATH-1121** Course Title: **Differential and Integral Calculus**

(Session: 2019-2020)

Time: **03** Hours Full Marks: **52.5**

[N.B. Answer 06 (Six) questions taking any 03 (Three) questions from each section. The figures in the right margin indicate full marks.]

Section-A

1. (a) Define a function, its domain and range with an example. 3

   (b) Sketch the graph of $f(x) = [x]$, where $[x]$ denotes the greatest integer not exceeding x, also find the domain and range of $f(x)$. 3

   (c) Define limit of a function $f(x)$ at $x = a$. 2.75

   Does $\lim_{x\to 0} f(x)$ exist for $f(x) = \begin{cases} 2x+1 & if\ x \le 0 \\ x^2 - x & if\ x > 0 \end{cases}$

2. (a) Define the continuity of a function $f(x)$ at a point $x = a$. Discuss the continuity 3

   of the function $f(x) = \begin{cases} x^2+1 & if\ x < 1 \\ 3 & if\ x = 1 \\ 4-2x & if\ x > 1 \end{cases}$ at $x = 1$.

   (b) Define the differentiability of a function at a point $x = a$. A function $f(x)$ is defined as follows: 3

   $$f(x) = \begin{cases} x & if\ 0 < x < 1 \\ 2-x & if\ 1 \le x \le 2 \\ x - \frac{x^2}{2} & if\ x \ge 2 \end{cases}$$

   Discuss the differentiability of the $f(x)$ at $x = 1$.

   (c) Find $\frac{dy}{dx}$ from $(cosx)^y = (siny)^x$. 2.75

3. (a) If $y = \sin(m \sin^{-1} x)$, then show that 2.75
   $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$.

   (b) State Rolle's Theorem. In the Mean Value Theorem $f(a+h) = f(a) + hf'(a+\theta h)$, 3
   if $a = 1, h = 3$ and $f(x) = \sqrt{x}$, find θ.

   (c) Examine whether $x^{\frac{1}{x}}$ possesses a maximum or a minimum and determine the same. 3

4. (a) Define a homogeneous function and state the Euler's Theorem on homogeneous functions. 2.75

   If $u = \tan^{-1}\frac{x^3+y^3}{x-y}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = sin2u$.

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University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engg. Part 1 Odd Semester Examination 2019

Course: MATH 1111 (Algebra, Trigonometry and Vector)

Full Marks: 52.5 Time: 3 Hours

[N.B. Answer any three questions from each Section]

Section-A

1)

a) Define a union and intersection of two sets. Let $A = \{0,1,2,3,4\}$ and $B = \{3,5,6\}$. Find $A \cup B$ and $A \cap B$. 2.75

b) Let X and Y be two sets. Define a relation and a function from X to Y. Let $X = \{1,2,3,4\}$ and $Y = \{a, b, c\}$. Consider the following three subsets of $X \times Y$ $f = \{(1, a), (2, a), (3, b), (4, c)\}$, $g = \{(2, a), (2, b), (3, b), (4, b)\}$ and $h = \{(1, a), (2, b), (3, c)\}$, which are functions from X into Y? Why? 3

c) Let $f: \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = 2x - 5$ and $g: \mathbb{R} \to \mathbb{R}$ defined by $g(x) = 5x^2 - 3$. Find fog and gof. 3

2)

a) Prove that imaginary roots of an equation occur in pairs (the equation with real co-efficients) . 3

b) Find the nature of the roots of the polynomial $f(x) = x^7 - x^4 - x^2 - 1$. 3

c) If a, b, c are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum a^2$. 2.75

3)

a) Solve the cubic equation $x^3 + x - 2 = 0$ 3

b) Prove that $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (b - c)(a - b)(c - a)(a + b + c)$ 3

c) Solve the following system of linear equations by using Cramer's rule: 2.75

$$x + y + z = 1, \quad x + 2y + z = 2, \quad x + y + 2z = 0.$$

4)

a) State De'Moivers Theorem. Find all the values of $(1 + i)^{1/3}$. 3

b) If $x + \frac{1}{x} = 2cos\theta$, then show that $x^n + \frac{1}{x^n} = 2cosn\theta$. 3

c) Prove that $i^i = e^{-(4n+1)\pi/2}$. 2.75

Section B

5)

a) Prove that 3

$$\frac{\Pi}{8} = \frac{1}{1.3} + \frac{1}{5.7} + \frac{1}{9.11} + \cdots$$

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Department of Computer Science and Engineering
B.Sc. Engg. Part-1 Odd Semester, Examination-2018
Course: MATH-1111 (Algebra, Trigonometry and Vector)

Time: 3 Hours | Full Marks: 52.5

(Answer any **six** questions taking **three** from each Section)

Section: A

1 (a) Define complement of a set. State and prove De' Morgan's rules — 3

(b) Evaluate $\begin{vmatrix} (b+c)^2 & c^2 & b^2 \\ c^2 & (c+a)^2 & a^2 \\ b^2 & a^2 & (a+b)^2 \end{vmatrix}$. — 3

(c) Solve the following system of linear equation by Cramer's rule: — 2.75
$x + y + z = 1;\ ax + by + cz = k;\ a^2x + b^2y + c^2z = k^2$

2 (a) State the fundamental theorem of algebra. Show that every equation of nth degree has exactly n roots. — 3

(b) Find the condition that $x^3+px^2+qx+r=0$ may have the roots in Harmonical progression. — 3

(c) If a, b, c are the roots of $x^3+qx+r=0$. Find the equation whose roots are $bc+\frac{1}{a}$, $ca+\frac{1}{b}$, $ab+\frac{1}{c}$. — 2.75

3 (a) Solve the cubic following equation by cardon's method: $x^3-21x-344=0$. — 3

(b) Show that the equation $\frac{A^2}{x-a}+\frac{B^2}{x-b}+\frac{C^2}{x-c}+\cdots+\frac{H^2}{x-h}=K$ has no imaginary roots. — 3

(c) If the roots of $x^n-1=0$ are $1, \alpha, \beta, \gamma \ldots$ Show that $(1-\alpha)(1-\beta)(1-\gamma)\ldots = n$. — 2.75

4 (a) State and prove De' Moivre's theorem. — 3

(b) If $x = \cos\theta + i\sin\theta$ and $1+\sqrt{1-a^2} = na$, prove that $1 + a\cos\theta = \frac{a}{2n}(1+nx)(1+\frac{n}{x})$. — 3

(c) If $x_r = cos\frac{\pi}{2^r} + isin\frac{\pi}{2^r}, r = 1,2,3,4,\ldots$ *Find the value of* $\prod_1^\infty x^r$. — 2.75

Section: B

5 (a) If $x = \frac{2}{1!}-\frac{4}{3!}+\frac{6}{5!}-\frac{8}{7!}+\cdots$ *to* ∞ *and* $y = 1+\frac{2}{1!}-\frac{2^3}{3!}+\frac{2^5}{5!}-\cdots$ *to* ∞, *show that* $x^2 = y$. — 3

(b) If $\tan\log(x+iy) = a+ib$ *where* $a^2+b^2 \neq 1$, *prove that* $\tan\log(x^2+y^2) = \frac{2a}{1-a^2-b^2}$. — 3

(c) Separate $\log(a+ib)$ into real and imaginary part. — 2.75

6 (a) If $sinx = nsin(a+x), -1 < n < 1$, *expand x in a series of ascending power of n.* — 3

(b) Given that $y = \log\tan\left(\frac{\pi}{4}+\frac{x}{4}\right) = x + c_3x^3 + c_5x^5 + \cdots$, *show that* $x = y - c_3y^3 + c_5y^5 - \cdots$ — 3

(c) Prove that $\frac{tan^{-1}x}{x}+\frac{tan^{-1}y}{y}+\frac{tan^{-1}z}{z} = 3[1-\frac{1}{7}+\frac{1}{13}-\frac{1}{19}+\frac{1}{25}-\cdots]$ where x, y, z are the three cube roots of unity. — 2.75

7 (a) Find the sum of the series $tan^{-1}\frac{1}{2.1^2}+tan^{-1}\frac{1}{2.2^2}+tan^{-1}\frac{1}{2.3^2}+\cdots$ *to n terms.* — 3

(b) Sum to n terms the series $tan\alpha + 2tan2\alpha + 2^2tan2^2\alpha + \cdots$ — 3

(c) Sum the series to infinity: $cos\theta - \frac{1}{2}cos2\theta + \frac{1}{3}cos3\theta - \cdots (-\pi < \theta < \pi)$. — 2.75

8 (a) If $\underline{r_1} = 2\underline{i} - \underline{j} + \underline{k}$, $\underline{r_2} = \underline{i} + 3\underline{j} - 2\underline{k}$, $\underline{r_3} = -2\underline{i} + \underline{j} - 3\underline{k}$, *and* $\underline{r_4} = 3\underline{i} + 2\underline{j} + 5\underline{k}$, find scalars a, b, c such that $\underline{r_4} = a\underline{r_1} + b\underline{r_2} + c\underline{r_3}$. — 3

(b) Prove that $\overline{\nabla}\times(\overline{\nabla}\times\overline{A}) = \overline{\nabla}(\overline{\nabla}.\overline{A}) - \nabla^2\underline{A}$. — 3

(c) A particle moves along the curve $x=2t^2$, $y=t^2-4t$, $z=3t-5$, where t is the time. Find the components of its velocity and acceleration at time t=1 in the direction $\underline{i} - 3\underline{j} + 2\underline{k}$ — 2.75

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc.in Engineering 1st Year 1st Semester Examination-2019
Course: **MATH 1211** [Differential and Integral Calculus]
Marks: 52.50 Time: 3 Hours
[Answer any six (06) questions taking three (03) from each section.]

Section-A

1.a) Define function and hence define domain codomain with an example. 3
b) Find the domain and range of $f(x) = |x+1| + |x|$ and also sketch the graph of $f(x)$. 3
c) Prove that every differential function is continuous but the converse is not true. 2.75

2.a) Define derivative of a function $f(x)$ at $x = c$. Show that the function 4

$$f(x) = \begin{cases} 3 + 2x \; for -\frac{3}{2} < x \leq 0 \\ 3 - 2x \; for \; 0 < x \leq \frac{3}{2} \end{cases}$$ is continuous at $x = 0$ but not differentiable at $x = 0$.

b) If $y = (\sin^{-1} x)^2$, then show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$. 3
c) i) Evaluate $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{\frac{1}{x}}$ ii) Differentiate $x^{\sin^{-1} x}$ with respect to $\sin^{-1} x$. 1.75

3.a) For a given curved surface of a right circular cone when the volume is maximum, show that the semi-vertical angle is $\sin^{-1}\frac{1}{\sqrt{3}}$. 4
b) Give the geometrical interpretation of Mean value theorem. 3
c) Verify Rolle's theorem for the function $f(x) = x^2 - 5x + 8$ in $[1, 4]$. 1.75

4.a) If $u = f(x^2 + 2xyz, y^2 + 2zx)$, prove that $(y^2 - zx)\frac{\partial u}{\partial x} + (z^2 - xy)\frac{\partial u}{\partial y} + (x^2 - yz)\frac{\partial u}{\partial z} = 0$. 4
b) Define pedal equation of a curve. Prove that the curve $\frac{x^2}{a} + \frac{y^2}{b} = 1$ and $\frac{x^2}{a'} + \frac{y^2}{b'} = 1$ will cut orthogonally if $a - b = a' - b'$. 3
c) Find the asymptotes of the curve $x^2y^2 - a^2(x^2 + y^2) = 0$. 1.75

Section-B

5. Answer any three of the following: 8.75
i) $\int \frac{x+1}{3+2x-x^2} dx$ ii) $\int \frac{dx}{\sqrt{(2x^2+3x+4)}}$
iii) $\int \frac{dx}{5+4\cos x}$ iv) $\int \frac{dx}{\sqrt{(x^2-2x+3)\;(x^2-2x+1)}}$

6.a) Evaluate $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$. 4
b) Evaluate $\int_2^4 \frac{x-3}{\sqrt{(5-x)(x-1)}} dx$. 3
c) Write down the five general properties of definite integral. 1.75

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B.Sc. Engg. Part-IOdd Semester, Examination-2017

Course: MATH1111 (Algebra, Trigonometry and Vector)

Time: 3 Hours **Marks: 52.5**

*(Answer **SIX(6)** questions taking **Three(3)** from each section)*

SectionA

1. (a) When the two sets A and B are said to be equal? Prove that $(A \cap B)' = A' \cup B'$. 2.75
   (b) Define an inverse function. Let V= {-2,-1, 0, 1, 2} and $g : V \to \Re^+$ be a function defined by the formula $f(x) = x^2 + 1$. Find the range of g. 3
   (c) Define a reflexive, symmetric and a transitive relation giving one example of each. 3

2. (a) State Descarte's rule of signs. Find the least possible number of imaginary roots of $x^9 + 5x^8 - x^3 + 7x + 2 = 0$. 3
   (b) Form the equation whose roots are the squares of the differences of the roots of $x^3 + qx + r = 0$. 2.75
   (c) Solve the cubic equation $x^3 - 15x = 126$ by Cardan's method. 3

3. (a) Prove that the imaginary root of an equation with real coefficient occur in pairs. 3
   (b) If a, b, c are roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum a^2$. 2.75
   (c) Using Cramer's rule solve the system of linear equations 3

$$2x - y - z = 4$$
$$x - 2y + z = 5$$
$$x - 2y + 2z = 1$$

4. (a) State De Moiver's theorem. Find the value of $(1+i)^{1/3}$. 3
   (b) If $x + \frac{1}{x} = 2\cos\theta$, then show that $x^n + \frac{1}{x^n} = 2\cos n\theta$. 3
   (c) Expand $\cos n\theta$ using DeMoiver's Theorem. 2.75

SectionB

5. (a) If $x = \log\tan(\frac{\pi}{4} + \frac{y}{2})$ then prove that $y = -i\log\tan(\frac{ix}{2} + \frac{\pi}{4})$. 3
   (b) Reduce $\tan^{-1}(\cos\theta + i\theta)$ to the form $a + ib$ and hence show that 3

$$\sin\theta - \frac{1}{3}\sin 3\theta + \frac{1}{5}\sin 5\theta - \cdots\cdots = \frac{1}{2}\log\{\pm\tan(\frac{\pi}{4} + \frac{\theta}{2})\}$$

   (c) If $\theta < \frac{\pi}{4}$, then prove that $\log\sec\theta = \frac{1}{2}\tan^2\theta - \frac{1}{4}\tan^4\theta + \frac{1}{6}\tan^6\theta - \cdots\cdots$. 2.75

6. (a) If S_n be the sum to n terms of the series $\sin x + \sin 2x + \sin 3x + \cdots\cdots$ then prove that 3

$$Lt_{n\to\infty} \frac{S_1 + S_2 + \cdots + S_n}{n} = \frac{1}{2}\cot\frac{x}{2}$$

   (b) Separate into real and imaginary parts the expression $\tan^{-1}(x + iy)$. 3
   (c) If $\tan(\alpha + i\beta) = x + iy$, Then prove that $x^2 + y^2 - 2y\coth 2\beta = -1$. 2.75

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8-A

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engineering **Part-I Odd** Semester Examination **2020**

Course Code: **MATH-1111** Course Title: **Algebra, Trigonometry and Vector**

(Session: **2017-2018** and **2018-2019**)

Time: **03** Hours Full Marks: **52.5**

[N.B. Answer 06 (Six) questions taking any 03 (Three) questions from each section. The figures in the right margin indicate full marks.]

Section-A

1. (a) Let A and B be two sets. Define $A - B$ and A'. Show that $A - B = A \cap B'$. 3

   (b) Show that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (b-c)(c-a)(a-b)$. 2.75

   (c) Solve the following system of linear equations by Cramer's rule: 3
   $x + y + z = 1, ax + by + cz = k, a^2x + b^2y + c^2z = k^2$.

2. (a) State the fundamental theorem of algebra. Show that every equation of nth degree has exactly n roots. 3

   (b) If a, b, c are the roots of the equation $x^3 - px^2 + qx - r = 0$, find the value of $\frac{1}{b^2c^2} + \frac{1}{c^2a^2} + \frac{1}{a^2b^2}$. 2.75

   (c) Show that the equation $\frac{A^2}{x-a} + \frac{B^2}{x-b} + \frac{C^2}{x-c} + \cdots + \frac{H^2}{x-h} = k$, has no imaginary roots. 3

3. (a) If the roots of the equation $x^n - 1 = 0$ are $1, \alpha, \beta, \gamma, \ldots$, then show that $(1-\alpha)(1-\beta)(1-\gamma)\ldots. = n$. 2.75

   (b) Form the equation whose roots are the squares of the differences of the roots of $x^3 + qx + r = 0$. 3

   (c) Solve the cubic equation $x^3 - 15x = 126$ by Cardan's method. 3

4. (a) Find the value of $(-1)^{1/6}$. 3

   (b) If $\left(x + \frac{1}{x}\right) = 2cos\theta$, then show that $x^n + \frac{1}{x^n} = 2cosn\theta$. 2.75

   (c) Expand $cos5\theta$ using De Moiver's Theorem. 3

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7. (a) Find the projection of the vector $A = \underline{i} - 2\underline{j} + \underline{k}$ on the vector $B = 4\underline{i} - 4\underline{j} + 7\underline{k}$. 2.75

(b) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\underline{i} - 3\underline{j} + 2\underline{k}$. 3

(c) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point (2,-1, 2). 3

8. (a) Prove that $\underline{\nabla} \times (\underline{\nabla} \times \underline{A}) = -\nabla^2 \underline{A} + \underline{\nabla}(\underline{\nabla}. \underline{A})$ 2.75

(b) Find the work done in moving a particle once around a circle c in the xy plane. If the circle has centre at the origin and radius 3 and if the force field is given by $\underline{F} = (2x - y + z)\underline{i} + (x + y - z^2)\underline{j} + (3x - 2y - 4z)\underline{k}$. 3

(c) Verify Stokes theorem for $\underline{A} = (2x - y)\underline{i} - yz^2\,\underline{j} - y^2 z\underline{k}$, where s is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and c is its boundary. 3

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